LLM-Guided Reasoning Diffusion Policy for Long-Horizon Manipulation

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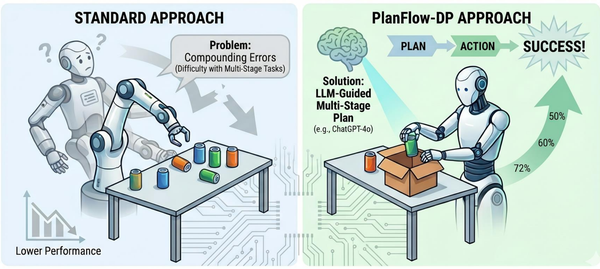
Introduction



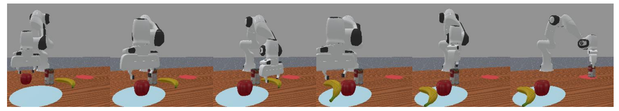
Long horizon manipulation is **HARD**!

Diffusion Policy alone is **NOT** enough!

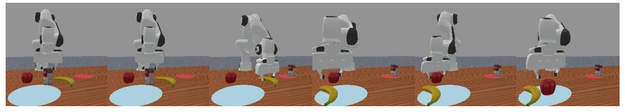
LLM are powerful **REASONERS**!

Task Setup

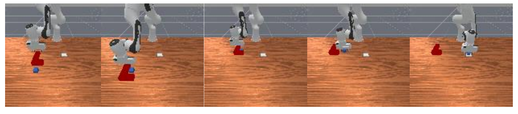
Test both **instruction following** and **complex long-horizon task** performance!



Sort Table: Put the fruits on the blue ellipse first, and then put the can on the red ellipse.

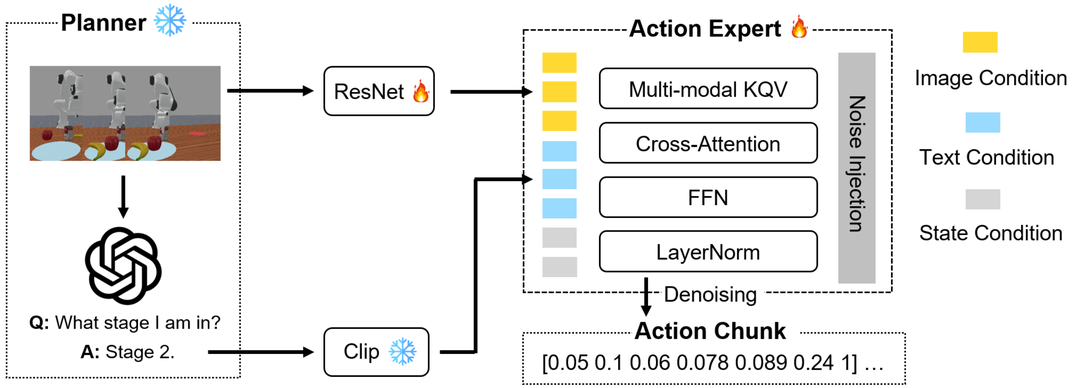


Sort Table Reverse: Put the can on the red ellipse first, and then put the fruits on the blue ellipse.

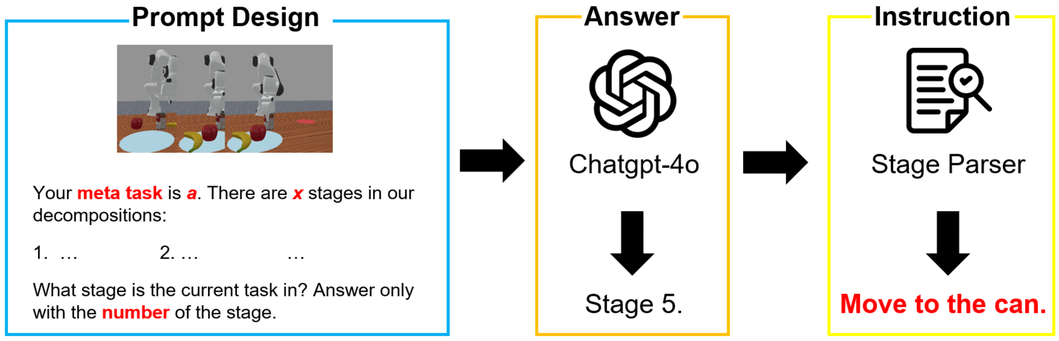


Tool Usage : Use the tool the reach the cube, and then put the cube into the bin.

Overall Structure



1 Stage-Aware Language Guidance



2 Diffusion-Based Action Prediction

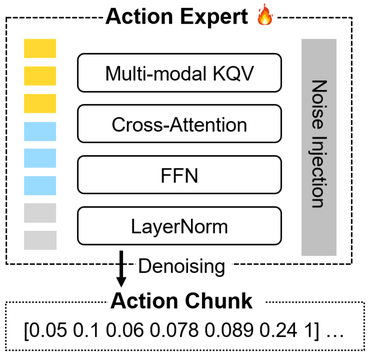


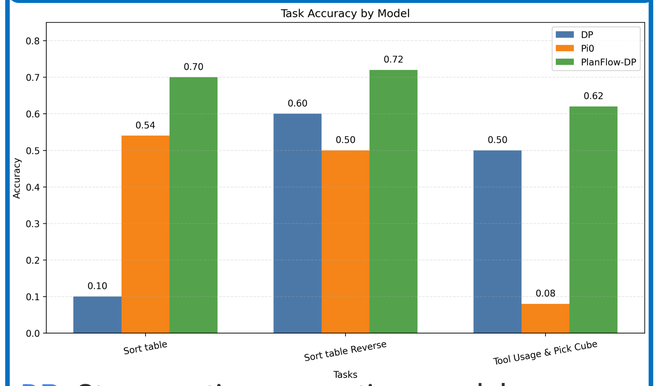
Image Condition — Inputs are **concatenated per step** and stacked over **obs_horizon**

Text Condition — State condition is the **robot joint position** vector.

State Condition — **Self-attention** is on **action tokens**; **cross-attention** is from **action tokens** to **condition tokens**.

Action is a **future action chunk** of **joint-position** control plus **gripper** command.

Results



DP: Strong action generation, weak language following. ☹

Pi0: Follows language but struggles on complex long-horizon tool-use tasks. ☹

PlanFlow-DP: Best on both language following and long-horizon complex tasks. ☺

Overall: We are the best!

Limitations

LLM hallucination: Strong action generation, weak language following.
Future: Fine-tune the LLM on robot data to improve planning accuracy.

Limited generalization: Large object-position shifts can cause policy failure!
Future: pretrain a larger policy on more robot data (Open X-Embodiment).

Control latency: API-based LLM calls introduce delay.
Future: use a local quantized fast-reasoning LLM to reduce latency.

References: [1] Nvidia, "R2d2: Perception-guided task & motion planning for long-horizon manipulation," 2025. [Online]. Available: https://developer.nvidia.com/blog/r2d2-perception-guided-task-amp-motion-planning-for-long-horizon-manipulation/
[2] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," 2024. [Online]. Available: https://arxiv.org/abs/2303.04137
[3] K. Black et al., "π0: A vision-language-action flow model for general robot control," 2026. [Online]. Available: https://arxiv.org/abs/2410.24164

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